



**TEMASEK
JUNIOR COLLEGE**

CHEMISTRY

8872/01

Paper 1 Multiple Choice

Wednesday

25 SEPTEMBER 2013

50 minutes

Additional materials: Data Booklet
 Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, index number and CG on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C, and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

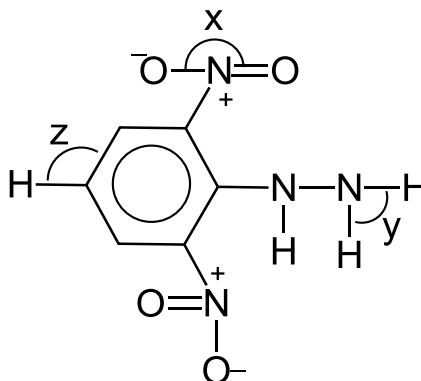
- 1 Which of the following formulae represents a particle with the composition 1 proton, 1 neutron and 2 electrons?
(D represents deuterium, ^2H)
- A** H
B D
C D^-
D He
- 2 *Use of the Data Booklet is relevant to this question.*
How many protons are there in 3.94 g of pure gold?
- A** 9.48×10^{22}
B 9.48×10^{23}
C 2.15×10^{23}
D 2.15×10^{24}
- 3 25 cm^3 sample of $0.100 \text{ mol dm}^{-3}$ YSO_4 solution required 10 cm^3 of $0.100 \text{ mol dm}^{-3}$ of acidified potassium manganate(VII) solution for complete reaction.
What is the change in oxidation state of element Y?
- A** +1
B +2
C +3
D +4
- 4 Carbon disulfide, CS_2 , is a poisonous liquid used in the manufacture of cellophane. It reacts with nitrogen monoxide, NO, to form sulfur and equal volumes of two gases. What are the two gases?
- A** CO, N_2
B CO, NO_2
C CO_2 , N_2
D CO_2 , NO_2

- 5 The first eight ionisation energies of an element **Q** (in kJ mol^{-1}) are as follows:

966, 1950, 2730, 4850, 6020, 12300, 15400, 18900

Which of the following statements about element **Q** is correct?

- A** It has an even number of protons in its atoms.
- B** It can form a chloride with the formula QCl_3 .
- C** It can form a chloride that is trigonal planar in shape.
- D** It is likely to have high melting point.
- 6 Which of the following pairs have the same shape?
- A** SO_2 , NO_2^+
- B** BF_3 , CO_3^{2-}
- C** PCl_3 , AlCl_3
- D** CH_4 , XeF_4
- 7 2,4-dinitrophenylhydrazine (Brady's reagent) is a common reagent used to qualitatively test for carbonyl groups associated with aldehydes and ketones. What would you expect the bond angles, x , y , and z to be, to the nearest 1 degrees, in the molecule of 2,4-dinitrophenylhydrazine?

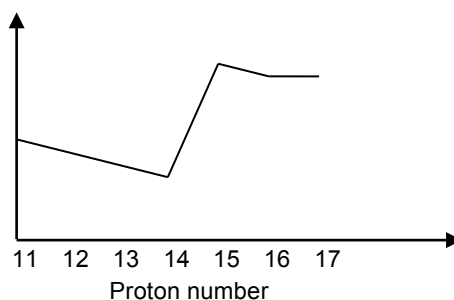


	X	Y	Z
A	107°	120°	120°
B	107°	107°	109°
C	109°	120°	120°
D	120°	107°	120°

- 8 Given that the lattice energies (in kJ mol^{-1}) of rubidium fluoride, RbF and caesium chloride, CsCl , are -760 and -650 respectively, which of the following values is likely to be the lattice energy of caesium fluoride, CsF ?

- A -500 kJ mol^{-1}
- B -630 kJ mol^{-1}
- C -720 kJ mol^{-1}
- D -820 kJ mol^{-1}

- 9 The graph shows how a property of the elements Na to Cl varies with proton number.



What is the property?

- A ionic radius
 - B first ionisation energy
 - C melting point
 - D electronegativity
- 10 A mixture of the oxide and chloride of elements in the third period is added to water. The resulting solution is acidic. What could be the constituents of the mixture?

- A Na_2O and AlCl_3
- B SiCl_4 and SiO_2
- C PCl_3 and Na_2O
- D Al_2O_3 and NaCl

- 11 The reaction between substances **A** and **B** is found to follow the rate law

$$\text{rate} = k[\text{A}]^m[\text{B}]$$

where k is the rate constant and has units of $\text{mol}^{-2} \text{dm}^6 \text{s}^{-1}$.

Two experiments to study the kinetics of this reaction were carried out and the data obtained are tabulated below:

Experiment	Initial [A] / mol dm^{-3}	Initial [B] / mol dm^{-3}	Initial rate / $\text{mol dm}^{-3} \text{s}^{-1}$
1	0.040	0.080	S
2	0.020	y	$S/2$

What is the value of y in experiment 2?

- A** 0.020
- B** 0.040
- C** 0.160
- D** 0.320
- 12 Iodine-131 is a radioactive isotope with a half-life of 8 days. Given that radioactive decay is a first-order reaction, what fraction of the isotope would remain after 48 days?

- A** $\frac{1}{2^4}$
- B** $\frac{1}{2^5}$
- C** $\frac{1}{2^6}$
- D** $\frac{1}{2^8}$

- 13** Hydrogen peroxide reacts with acidified iodide ions, liberating iodine. In investigations of this reaction, the following results were obtained.

initial concentration of reactants / mol dm ⁻³			initial rate of formation of iodine / mol dm ⁻³ s ⁻¹
[H ₂ O ₂]	[I ⁻]	[H ⁺]	
0.010	0.010	0.10	2 x 10 ⁻⁶
0.030	0.010	0.10	6 x 10 ⁻⁶
0.030	0.020	0.10	1.2 x 10 ⁻⁵
0.030	0.020	0.20	1.2 x 10 ⁻⁵

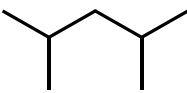
Which of the following statements follows from these results?

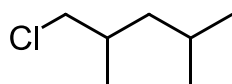
- A** The overall order of this reaction is 2.
- B** The unit of the rate constant k is mol⁻² dm⁶ s⁻¹.
- C** This reaction can be catalysed by H⁺.
- D** The rate equation for the reaction can be written as rate = $k[\text{H}_2\text{O}_2]^2 [\text{I}^-]$.
- 14** Two equilibria are shown below.
- Reaction I: $2\text{NO}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}) + \text{O}_2(\text{g})$
- Reaction II: $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
- The numerical value K_c for reaction I is 0.36. Under the same conditions, what is the numerical value for K_c for reaction II?
- A** 0.18
- B** 1.7
- C** 2.8
- D** 5.6

- 15 An enzyme operates at maximum efficiency when in an aqueous solution buffered at pH 9. Which combination of substances, when dissolved in 2 dm³ of water, would give the necessary buffer solution?

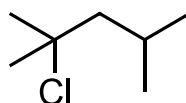
- A 1 mol of CH₃CO₂NH₄
 B 1 mol of NH₄Cl and 1 mol of NH₃
 C 1 mol of HCl and 1 mol of CH₃COOH
 D 1 mol of CH₃COOH and 1 mol of CH₃CO₂Na

16

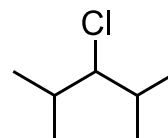
When 2,4-dimethylpentane, , is subjected to Cl₂(g) under suitable conditions, a mixture of products are obtained as follows:



product 1



product 2



product 3

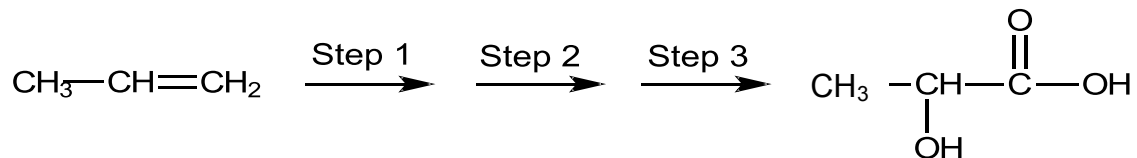
What is the ratio of the products formed?

	product 1	product 2	product 3
A	3	1	1
B	3	2	2
C	6	1	1
D	6	2	1

- 17 Which of the following shows the correct trend of increasing boiling point?

- A CH₃I, CH₃Br, CH₃Cl
 B CH₃CH₂CH₂NH₂, CH₃CH₂CH₂OH, HOCH₂CH₂OH
 C CH₃CH₂CH₂CH₃, CH₃CH₂CH₂OH, CH₃COCH₃
 D CH₃(CH₂)₃CH₃, (CH₃)₂CHCH₂CH₃, CH₃C(CH₃)₂CH₃

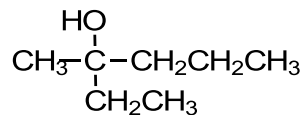
- 18** Lactic acid, also known as milk acid, is a chemical compound that plays a role in various biochemical processes. Lactic acid can be synthesised from propene in a 3-step process.



Which of the following will give the highest yield of lactic acid?

	Step 1	Step 2	Step 3
A	Br ₂ (g), room temp	NaOH(aq), warm	Cr ₂ O ₇ ²⁻ / H ⁺ , warm
B	Br ₂ (aq), room temp	Cr ₂ O ₇ ²⁻ / H ⁺ , warm	NaOH(aq), warm
C	MnO ₄ ⁻ / H ⁺ , cold	Cr ₂ O ₇ ²⁻ / H ⁺ , warm	H ₂ / Pt, room temp
D	MnO ₄ ⁻ / OH ⁻ , cold	Cr ₂ O ₇ ²⁻ / H ⁺ , warm	LiAlH ₄ , room temp

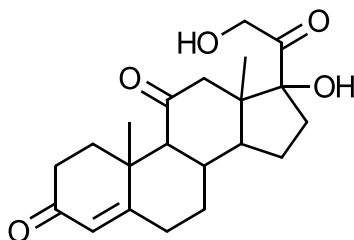
- 19** How many different isomers can be formed when excess concentrated sulfuric acid at 180 °C?



is heated with

- A** 2
B 3
C 4
D 5

- 20 Cortisone is one of the main hormones that is released in response to stress.



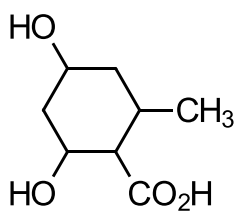
Which of the following is correct?

- A It exhibits geometric isomerism.
- B It reacts with 4 moles of ethanolic NaBH_4 .
- C Effervescence is observed when $\text{Na}_2\text{CO}_3(\text{s})$ is added.
- D It gives off white fumes when refluxed with SOCl_2 in pyridine.
- 21 Which of the following **cannot** be used to synthesise pentanoic acid in not more than 2 steps?
- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OCOCH}_2\text{CH}_3$
- B $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)_2\text{OH}$
- C $\text{CHOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
- D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
- 22 Which of the following shows the correct order of increasing pK_a ?
- A $\text{CH}_3\text{CO}_2\text{H} < \text{CH}_3\text{CH}_2\text{OH} < \text{H}_2\text{O}$
- B $\text{CH}_3\text{CH}_2\text{CO}_2\text{H} < \text{CH}_3\text{CO}_2\text{H} < \text{CH}_3\text{CH}_2\text{OH}$
- C $\text{C}/\text{CH}_2\text{CO}_2\text{H} < \text{Cl}_2\text{CHCO}_2\text{H} < \text{Cl}_3\text{CCO}_2\text{H}$
- D $\text{CH}_3\text{CH}_2\text{CCl}_2\text{CO}_2\text{H} < \text{CH}_3\text{CHClCHClCO}_2\text{H} < \text{CH}_2\text{ClCH}_2\text{CHClCO}_2\text{H}$

- 23 Which of the following reagents and conditions **cannot** be used to distinguish the pair of compounds in each set?

	Compound 1	Compound 2	Reagent and condition
A	$\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_3$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$	$[\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$, warm
B	$\text{CH}_3\text{CH}_2\text{COCHI}_2$	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$	$\text{I}_2(\text{aq})$, $\text{NaOH}(\text{aq})$, warm
C	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$	$\text{AgNO}_3(\text{aq})$, room temperature
D	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$	$\text{K}_2\text{Cr}_2\text{O}_7(\text{aq}) / \text{H}^+$, Hot

24



Compound **Y**

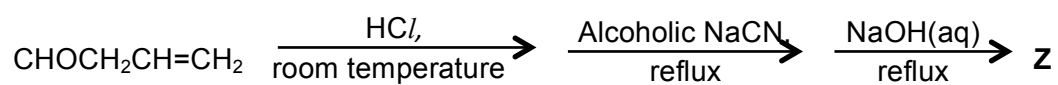
0.1 mol of compound **Y** was reacted with excess $\text{Na}_2\text{CO}_3(\text{s})$ and the gaseous product formed was passed through a bottle of excess concentrated NaOH .

What is the increase in mass in the bottle of concentrated NaOH ?

- A** 2.2 g
- B** 4.4 g
- C** 6.6 g
- D** 13.2 g

25

Compound **Z** is formed from the following reaction pathway:



What is the most likely structural formula of the organic product formed?

- A** ${}^+\text{Na}^-\text{O}_2\text{CCH}(\text{OH})\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$
- B** $\text{HOCH}(\text{CN})\text{CH}_2\text{CH}_2\text{CH}_2\text{CN}$
- C** $\text{CHOCH}_2\text{CH}(\text{CH}_3)\text{CO}_2^-\text{Na}^+$
- D** $\text{CHOCH}_2\text{CH}_2\text{CH}_2\text{OH}$

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

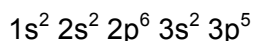
Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26** **Y** has the following electronic configuration.



What could **Y** be?

- 1** Cl
- 2** S⁻
- 3** Ar⁺

- 27** Which reactions represent standard enthalpy changes at 298 K?

- 1** $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$
- 2** $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
- 3** $\text{C}(\text{s}) + 8\text{H}(\text{g}) \rightarrow \text{C}_3\text{H}_8(\text{g})$

- 28** Element **E** is in the third period of the Periodic Table. The chloride of **E** has a simple molecular structure while the oxide of **E** has a giant ionic structure. Which of the following statements are correct?

- 1** The atomic radius of **E** is larger than that of chlorine.
- 2** The second ionisation energy of **E** is lower than that of sodium.
- 3** The oxide of **E** does not react with excess aqueous sodium hydroxide.

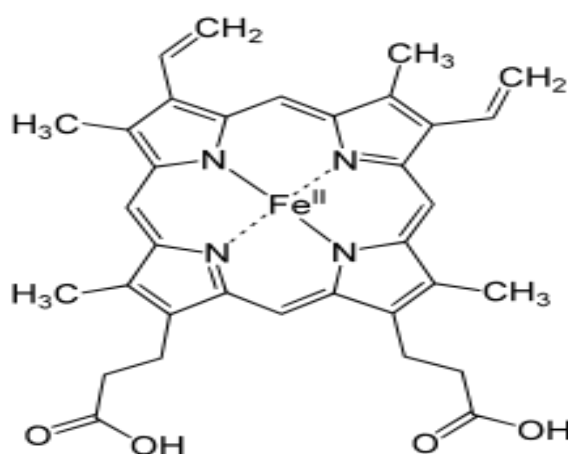
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 29** Haemoglobin has a quaternary structure with four globular protein subunits. Each subunit of haemoglobin has an embedded heme group which contains one iron atom that can bind to one oxygen molecule.

The diagram below shows the heme group held in a heterocyclic ring, known as the porphyrin.



Note that the four N atoms and the Fe ion are planar.

Which of the following descriptions of the bonds are present between Fe and the N-atoms in the porphyrin ring?

- 1** Ion-dipole interactions
- 2** Ionic Bonds
- 3** Coordinate bonds

The responses **A** to **D** should be selected on the basis of

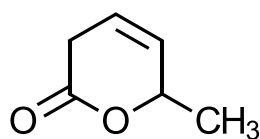
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

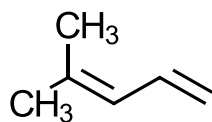
- 30** Compound **X** forms 2 organic products when refluxed with acidified KMnO_4 . One of the products can react with aqueous alkaline I_2 to give a yellow precipitate.

Which of the following could **X** be?

1



2



3

